Balanced search trees - synthetic teaching deck

Page 1: Binary search and balance

A binary search tree keeps smaller keys on the left.

Larger keys are on the right.

The balance factor is height(left) - height(right).

An AVL tree keeps this factor in {-1, 0, 1}.

Page 2: Right rotation

Start: root 30, left child 20, left grandchild 10.

The inorder sequence is 10, 20, 30.

Rotate right around 30: 20 becomes the new root.

10 is its left child and 30 its right child.

A rotation preserves the inorder sequence.

Page 3: Why balance matters

An AVL tree has height O(log n).

Search takes O(log n) time in a balanced tree.

Insertion may require rotations to restore balance.

Static definitions do not need an animation.

Verify every answer against this synthetic deck.